MIL-C-12280A(CmlC)
29 December 1955
SUPERSEDING
MIL-C-12280(CmlC)
20 October 1952

MILITARY SPECIFICATION

CARTRIDGE, ADSORBENT, M2

1. SCOPE

1.1 This specification covers an adsorbent cartridge for use as a component of a respirator for protection against paint vapors.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-621 - Boxes, Wood, Nailed and Lock-Corner.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

FSC 4240

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DRAWINGS

CHEMICAL CORPS

B21-33-131 - Fixture, Adsorbent Cartridge - Bill of Material.
B21-36-27 - Test Fixture (For Cartridge, Adsorbent, M2) - Bill of Material.
C5-20-96 - Packing, Adsorbent Cartridge - Bill of Material.
C5-71-2 - Cartridge, Adsorbent, M2 - Bill of Material.

PUBLICATIONS

CHEMICAL CORPS PAMPHLET

Pamphlet No. 2

Part I, Section A - Canister Test Methods - Chlorpicrin.
Part II, Section L - Adsorbent Test Methods - Analytical Test Methods.

CHEMICAL CORPS DIRECTIVES

110 - Operation of Chemical Container Volume Meter for Gas Mask Canisters.
112 - Operation of Canister, Air Purifier and Chemical Container Resistance Indicator, E3 Used in the Testing of Gas Mask Canisters and Filter Material.
380 - Operation of Machine, Rough Handling, E5.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED FREIGHT CLASSIFICATION COMMITTEE

Consolidated Freight Classification No. 20.

(Copies may be obtained from the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Material. Materials used in the manufacture of the cartridge shall

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conform to the specifications, standards, and drawings forming a part of this specification.

3.1.1 Filling.

3.1.1.1 Volume. The cartridge shall contain the charcoal specified on Drawing C5-71-3 which is listed on Drawing C5-71-2. Each cartridge shall contain 110 ± 5 milliliters of charcoal and the surface of the charcoal shall be level at the end of the filling operation (see 6.3).

3.1.1.2 Moisture content. The moisture content of the charcoal in each cartridge shall be not more than 5.0 percent by weight when tested as specified in 4.4.2.

3.2 Assembly. The cartridge shall be assembled, weighed to the nearest gram and marked as specified on Drawing C5-71-3 which is listed on Drawing C5-71-2.

3.3 Air-flow resistance. The air-flow resistance of the assembled cartridge measured as a pressure differential, shall be 45 ± 5 millimeters of water when tested as specified in 4.4.3.

3.4 Rough handling. The cartridge shall show no breaks or open seams when tested as specified in 4.4.4.

3.5 Gas life. The gas life of the cartridge shall be less than 30 minutes when tested as specified in 4.4.5.

3.6 The package container (see 5.1) shall not leak when tested as specified in 4.5 (see 6.4).

3.7 Workmanship. The cartridge shall be free from any damage, defects, or contamination.

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot. A lot shall consist of the cartridges produced by one manufacturer, from the same materials, and under essentially the same manufacturing conditions. However, not more than one lot of charcoal shall be represented in any one lot of finished cartridges. The inspection lot for nondestructive inspection, within the above limitations, shall be designated by the inspector.

4.2 Sampling.

4.2.1 For nondestructive inspection. Sampling for nondestructive inspection shall be conducted in accordance with the classification of defects (4.3.3) and Standard MIL-STD-105.

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4.2.2 For tests. Eighteen sealed containers, nine for test and nine for retest, shall be taken at random from each lot and shall be tested as indicated in table I (see 6.2).

Table I. Samples for tests

	No. of samples (cartridges) (36)		
Requirement	Test	Retest	Test method
Charcoal volume	)	)	4.4.1
Moisture content*	) 9	) 9	4.4.2
Rough handling	)	)	4.4.4
Gas protection**	) 9	) 9	4.4.5

*Use the material from the cartridges tested for charcoal volume.
**Use cartridges which have been subjected to the rough handling test.

4.3 Inspection.

4.3.1 Components. The inspector shall ascertain that all materials, components, or subassemblies of the cartridge covered by separate drawings, specifications, or standards are in accordance with the applicable drawings, specifications, or standards prior to final acceptance of the cartridge.

4.3.2 Packaging, packing, and marking. The inspector shall inspect the packaging, packing, and marking for compliance with section 5.

4.3.3 Classification of defects. Inspection of the cartridge shall be conducted as follows:

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4.3.3.1 Cartridge, adsorbent, M2. (Dwg. C5-71-3 which is listed on Dwg. C5-71-2)

Categories and defects	Acceptance standards
Major A	AQL 0.25 percent defectives
1a. Air-flow resistance	4.4.3
Major	AQL 1.0 percent defectives
1. Component missing	
2. Diameter incorrect	
3. Damage	
4. Marking incorrect	
Minor	AQL 2.5 percent defectives
51. Improper seaming	
52. Contamination	
53. Coating missing or inadequate	
54. Marking missing ir illegible	
Incidental	
None	

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4.3.3.2 Packaging, adsorbent cartridge. (Dwg. C5-20-92 which is listed on Dwg. C5-20-96)

Categories and defects	Acceptance standards
Major A	AQL 0.65 percent defectives
1a. Leakage	4.5
Major	AQL 2.5 percent defectives
1. Marking incorrect	
2. Key missing	
3. Damage	
4. Scoring missing	
Minor	AQL 4.0 percent defectives
51. Marking missing or illegible	
52. Cartridge loose in container	
53. Coating missing or inadequate	
54. Improper seaming	
Incidental	AQL 10 percent defectives
101. Contamination	

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4.4 Tests.

4.4.1 Charcoal volume. Determine the charcoal volume of the cartridge in accordance with Directive 110.

4.4.2 Moisture content. Determine the moisture content of the charcoal in the cartridge in accordance with Pamphlet No. 2, Part II, Section M.

4.4.3 Air-flow resistance. Determine the air-flow resistance of the cartridge at 85 liters of air per minute as specified in Directive 112. Use the fixture shown on Drawing C21-36-28 which is listed on Drawing B21-36-27.

4.4.4 Rough handling. Rough handle the cartridges for 30 minutes in accordance with Directive 380 and examine for compliance with 3.4.

4.4.5 Gas life. Determine the gas life of the rough handled cartridge in accordance with Pamphlet No. 2, Part I, Section A, at the following conditions using fixture shown on Drawing B21-33-132 which is listed on Drawing B21-33-131.

Chloropicrin concentration	-	10 milligrams per liter
Flow rate	-	32 liters per minute
Relative humidity	-	50 percent
Temperature	-	25° ∠ 5°C.

4.5 Container leakage. After closing, the container (see 5.1) shall be submerged not more than 2 inches below the surface of water at 85° ∠ 2°C. for 30 seconds and observed for leakage (see 6.4).

4.6 Acceptance criteria. If one specimen cartridge fails in any of the tests, specified in Table I, retests shall be made, using the specimens taken for this purpose (see 4.2.2). If more than one specimen fails in the test and retest combined, the lot shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Two cartridges shall be packaged in a hermetically sealed metal container as specified on Drawing C5-20-92.

5.1.1 Level A, military package. Fifty sealed containers (100 cartridges) packaged as in 5.1 shall be packaged in a fiberboard box as specified on Drawing C5-20-98.

5.1.2 Level B, limited military package. Same as 5.1.1.

5.1.3 Level C, minimum military package. Not applicable.

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5.2 Packing.

5.2.1 Level A, military pack. Two fiberboard containers (200 cartridges) packaged as in 5.1.1 shall be packed in a wood box as specified on Drawing D5-20-97.

5.2.2 Level B, limited military pack. Same as 5.2.1 except that the nailed wood box and wood thickness shall be in accordance with Specification NN-B-621.

5.2.3 Level C, minimum military pack. One hundred sealed containers (200 cartridges) packaged as in 5.1 shall be packed in a fiberboard box conforming to the Uniform Freight Classification.

5.3 Marking.

5.3.1 Metal container. The metal container holding two cartridges shall be marked as shown on Drawing D5-20-93.

5.3.2 Fiberboard container. The fiberboard container holding 100 cartridges shall be marked as shown on Drawing C5-20-98.

5.3.3 The exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification.
b. Type of packing required (see 5.2).

6.2 Sampling and testing. When the contractor consistently produces high quality material and operates under a system of quality control acceptable to the Government, the Government, at its discretion, may modify in whole or in part, the sampling and testing procedures specified herein. However, the Government reserves the right to return at any time, without previous notice to the contractor, to the sampling and testing procedures specified in this specification.

6.3 Consolidation pressure. Cartridges which have had charcoal consolidated under a load of 350 ± 25 pounds have been found to meet the requirements of this specification. This information is furnished as an aid to manufacture and is not a criterion for consolidation.

6.4 Leaking containers. Containers showing leakage in the test specified in 4.5 may be reconditioned, by wiping dry and sealing the leak with solder or other suitable means, and retested.

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